

KK Cryptosystem — Complete Mathematical Formulation

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1 Constants & Initialisation Vector

$$N = 25, \quad R = 32, \quad R_{\text{kdf}} = 20$$

$$W_{\text{rate}} = 19, \quad W_{\text{cap}} = 6, \quad B_{\text{rate}} = 8 \cdot W_{\text{rate}} = 152$$

$$\delta_H = 0\text{x}01, \quad \delta_K = 0\text{x}02, \quad \delta_M = 0\text{x}03$$

$$\text{IV}[i] = \left\lfloor \frac{\text{frac}(\sqrt{p_i}) \cdot 2^{64}}{1} \right\rfloor, \quad i \in \{0, \dots, 24\}, \quad p_i = i\text{-th prime}$$

$$\rho = (\rho_0, \rho_1, \dots, \rho_{14}) \in (\{1, 3, \dots, 63\}^2)^{15} \quad (15 \text{ rotation pairs, all odd})$$

$$D_k = \text{DIAGS}[k] \subset \{0, \dots, 24\}, \quad |D_k| = 5, \quad k \in \{0, \dots, 4\}$$

2 Core Primitives

MFR (Multiply-Fold-Rotate):

$$\text{MFR}(a, b, r) : \quad p = a \cdot (b \mid 1) \bmod 2^{64}, \quad \text{MFR} = \text{ROL}_r(p \oplus (p \gg 32))$$

DDR (Data-Dependent Rotation):

$$\text{DDR}(a, b) : \quad f = b \oplus (b \gg 32), \quad s = (f \oplus (f \gg 16) \oplus (f \gg 8)) \& 63, \quad \text{DDR} = \text{ROL}_s(a)$$

Quintet round on words (a, b, c, d, e) with rotation pair (ρ_0, ρ_1) :

$$a \leftarrow \text{MFR}(a, b, \rho_0)$$

$$c \leftarrow c \oplus a$$

$$d \leftarrow \text{DDR}(d, c)$$

$$e \leftarrow \text{MFR}(e, d, \rho_1)$$

$$b \leftarrow b \oplus e$$

3 Permutation π

For round $r \in \{0, \dots, R-1\}$, state $S \in \{0, 1\}^{64 \times 25}$ on a 5×5 grid:

Phase 1 — Rows ($j = 0 \dots 4$):

$$(S[5j], S[5j+1], S[5j+2], S[5j+3], S[5j+4]) \leftarrow \text{Quintet}(\dots, \rho_j)$$

Phase 2 — Columns ($j = 0 \dots 4$):

$$(S[j], S[j+5], S[j+10], S[j+15], S[j+20]) \leftarrow \text{Quintet}(\dots, \rho_{5+j})$$

Phase 3 — Diagonals ($k = 0 \dots 4$):

$$(S[D_k[0]], \dots, S[D_k[4]]) \leftarrow \text{Quintet}(\dots, \rho_{10+k})$$

Round-constant injection (grid corners + centre):

$$S[0] \text{ += } r, \quad S[4] \text{ += } r \cdot \phi, \quad S[12] \text{ += } r \cdot e', \quad S[20] \text{ += } r \cdot \pi', \quad S[24] \text{ += } r \cdot c_4$$

where $\phi = 0x9E3779B97F4A7C15$, $e' = 0xB7E151628AED2A6A$, $\pi' = 0x243F6A8885A2F7A4$, $c_4 = 0x298B075B4B6A5240$, all mod 2^{64} .

Re-keying (when $S \bmod 8 = 7$):

$$S[i] \leftarrow S[i] \oplus \text{ROL}_r(S[W_{\text{rate}} + (i \bmod W_{\text{cap}})]), \quad i \in \{0, \dots, W_{\text{rate}} - 1\}$$

4 Sponge Construction

Initialise: $S \leftarrow \text{IV}$, rotations ρ

Absorb block B (zero-padded to W_{rate} words):

$$S[i] \leftarrow S[i] \oplus B_i \quad (i = 0 \dots W_{\text{rate}} - 1), \quad S \leftarrow \pi_R(S)$$

Finalize with domain byte δ :

$$\text{pad} = [\delta, 0x80, 0, \dots, 0, 0x01] \quad (B_{\text{rate}} \text{ bytes}), \quad \text{Absorb}(\text{pad})$$

Squeeze n bytes (full $R = 32$ rounds):

$$\text{output} \leftarrow \text{output} \parallel S[0 \dots W_{\text{rate}} - 1], \quad S \leftarrow \pi_R(S), \quad \text{repeat until } n \text{ bytes emitted}$$

Squeeze-KDF n bytes (reduced $R_{\text{kdf}} = 20$ rounds):

$$\text{Same, with } S \leftarrow \pi_{R_{\text{kdf}}}(S)$$

5 Hash, MAC, KDF

$$\text{KK-Hash}(M) : \quad \text{Absorb}(M) \rightarrow \text{Finalize}(\delta_H) \rightarrow \text{Squeeze}(32)$$

$$\text{KK-MAC}(K, M) : \quad \text{Absorb}(K) \rightarrow \text{Absorb}(M) \rightarrow \text{Finalize}(\delta_M) \rightarrow \text{Squeeze}(32)$$

$$\text{KK-MAC}_\varepsilon(K, M, \varepsilon) : \quad \text{Absorb}(K) \rightarrow \text{Absorb}(\varepsilon) \rightarrow \text{Absorb}(M) \rightarrow \text{Finalize}(\delta_M) \rightarrow \text{Squeeze}(32)$$

$$\text{KK-KDF}(s, \text{salt}, \text{info}, n) : \quad \text{Absorb}(s) \rightarrow \text{Absorb}(\text{salt}) \rightarrow \text{Absorb}(\text{info}) \rightarrow \text{Finalize}(\delta_K) \rightarrow \text{Squeeze-KDF}(n)$$

6 Entropy

Gather $\rightarrow \varepsilon = (\varepsilon_b, \varepsilon_t)$ where $\varepsilon_b \in \{0, 1\}^{256}$, $\varepsilon_t \in \{0, 1\}^{128}$:

$$s_0 = \text{OsRng}(32), \quad s_1 = \text{timestamp_nanos_LE}(16)$$

$$s_2 = (\text{rdtsc} \oplus \text{stack_addr})_{\text{LE}(8)}, \quad s_3 = \text{KK-Hash}(\text{jitter}_{64})(32)$$

$$\varepsilon_b = \text{kk_entropy_mix}(\{s_0, s_1, s_2, s_3\}, 32), \quad \varepsilon_t = \text{timestamp at capture}$$

kk_entropy_mix $(\{s_0, \dots, s_{m-1}\}, n)$:

$$\text{Init sponge } S, \quad \text{for } i = 0 \dots m-1 : \text{Absorb}(i_{\text{LE}(8)} \parallel \text{len}(s_i)_{\text{LE}(8)} \parallel s_i)$$

$$\text{Finalize}(\delta_H) \rightarrow \text{Squeeze}(n)$$

Serialise: $\text{ser}(\varepsilon) = \varepsilon_b \parallel (\varepsilon_t)_{\text{LE}(16)} \quad (48 \text{ bytes})$

7 Symbol & Commitment Key Derivation

$$K_{\text{sym}}(K, j, t) = \text{KK-KDF}\left(K, \emptyset, \text{"KK-sym-v1"} \parallel 0x00 \parallel j_{\text{LE}(8)} \parallel t_{\text{LE}(16)}, B_{\text{rate}}\right)$$

$$K_{\text{commit}}(K, \varepsilon) = \text{KK-KDF}(K, \varepsilon_b, \text{"KK-commit-v1"}, 32)$$

Batch (8 consecutive indices, vectorised):

$$(K_{\text{sym}}^{(j)}, \dots, K_{\text{sym}}^{(j+7)}) = \text{KK-KDF-Batch}_8(K, j, t)$$

8 XOR Keystream Encryption

$$\text{CHUNK} = 4096, \quad \text{BATCH} = 8 \cdot \text{CHUNK} = 32768$$

$$P = P_0 \| P_1 \| \dots \| P_{\lceil L/\text{CHUNK} \rceil - 1}$$

Full 8-chunk batches (parallel):

$$(K_{\text{sym}}^{(8i)}, \dots, K_{\text{sym}}^{(8i+7)}) \leftarrow \text{Batch}_8(K, 8i, \varepsilon_t)$$

$$C_{8i+k} = P_{8i+k} \oplus K_{\text{sym}}^{(8i+k)}[0 \dots |P_{8i+k}| - 1], \quad k = 0 \dots 7$$

Tail chunks (scalar):

$$K_{\text{sym}}^{(j)} \leftarrow K_{\text{sym}}(K, j, \varepsilon_t), \quad C_j = P_j \oplus K_{\text{sym}}^{(j)}[0 \dots |P_j| - 1]$$

9 Temporal Commitment

Standard:

$$K_c = K_{\text{commit}}(K, \varepsilon), \quad \tau = \text{KK-MAC}(K_c, \varepsilon_b \| (\varepsilon_t)_{\text{LE}} \| C)$$

AEAD:

$$\tau = \text{KK-MAC}(K_c, |A|_{\text{LE}(8)} \| A \| \varepsilon_b \| (\varepsilon_t)_{\text{LE}} \| C)$$

Bound (temporal proof, 96 bytes):

$$\tau = \text{KK-MAC}_\varepsilon(K_c, \nu \| \tau_{\text{prev}} \| \varepsilon_b \| (\varepsilon_t)_{\text{LE}} \| C, \varepsilon_b)$$

$$\text{TemporalProof} = \nu(32) \| \varepsilon_b(32) \| \tau(32)$$

10 Encode / Decode

Encode (K, P):

$$\begin{aligned} \varepsilon &\leftarrow \text{Gather}(), \quad C \leftarrow P \oplus \text{Keystream}(K, \varepsilon), \quad \tau \leftarrow \text{Commit}(K, \varepsilon, C) \\ &\text{return } (\text{ser}(\varepsilon), C, \tau) \end{aligned}$$

Decode ($K, \text{ser}(\varepsilon), C, \tau$):

$$\text{verify } \tau, \quad P \leftarrow C \oplus \text{Keystream}(K, \varepsilon)$$

Encode-AEAD (K, P, A): same with $\text{Commit}_{\text{aead}}$

Encode-Bound ($K, P, \nu, \tau_{\text{prev}}$): same with $\text{Commit}_{\text{bound}}$

11 KkRng — Deterministic Generator

$$\text{Init}(\text{seed}) : \quad \sigma_0 = \text{KK-Hash}(\text{seed})$$

$$\text{Next}(n) : \quad \text{out} = \text{KK-KDF}(\sigma_i, \emptyset, \text{"KkRng-next"}, n), \quad \sigma_{i+1} = \text{KK-Hash}(\sigma_i)$$

$$\text{Reseed}(z) : \quad \sigma \leftarrow \text{KK-Hash}(\sigma \| z)$$

12 EKA — Ephemeral Key Agreement (3-message, PSK)

$$\varepsilon_A \leftarrow \text{Gather}(), \quad \varepsilon_B \leftarrow \text{Gather}()$$

Message 1 (Alice $\rightarrow$ Bob, 32 B):

$$m_1 = \text{KK-Hash}(\text{ser}(\varepsilon_A))$$

Message 2 (Bob $\rightarrow$ Alice, 80 B):

$$m_2 = \text{ser}(\varepsilon_B) \parallel \text{KK-MAC}(\text{psk}, \text{ser}(\varepsilon_B) \parallel m_1)$$

Message 3 (Alice $\rightarrow$ Bob, 80 B):

$$m_3 = \text{ser}(\varepsilon_A) \parallel \text{KK-MAC}(\text{psk}, \text{ser}(\varepsilon_A) \parallel \text{ser}(\varepsilon_B))$$

Bob verifies:

$$\text{KK-Hash}(\text{ser}(\varepsilon_A)) \stackrel{?}{=} m_1 \quad \wedge \quad \text{verify auth}_A$$

Session key:

$$K_{\text{sess}} = \text{KK-KDF}(\text{psk}, \text{ser}(\varepsilon_A) \parallel \text{ser}(\varepsilon_B), \text{“KK-EKA-session”, } 32)$$

13 ROPE — 4-Strand Ratchet

Init (K, ctx):

$$\text{salt} = \text{KK-Hash}(\text{ctx})$$

$$\sigma_{\text{ent}} = \text{KK-KDF}(K, \text{salt}, \text{“KK-rope-init-ent”, } 32)$$

$$\sigma_{\text{tmp}} = \text{KK-KDF}(K, \text{salt}, \text{“KK-rope-init-tmp”, } 32)$$

$$\sigma_{\text{chn}} = \text{KK-KDF}(K, \text{salt}, \text{“KK-rope-init-chn”, } 32)$$

$$n = 0$$

Step (ε):

$$\sigma'_{\text{ent}} = \text{KK-KDF}(\sigma_{\text{ent}}, \varepsilon_b, \text{“KK-rope-ent-v1”, } 32)$$

$$\sigma'_{\text{tmp}} = \text{KK-KDF}(\sigma_{\text{tmp}}, (\varepsilon_t)_{\text{LE}(16)}, \text{“KK-rope-tmp-v1”, } 32)$$

$$n \leftarrow n + 1, \quad \sigma'_{\text{chn}} = \text{KK-KDF}(\sigma_{\text{chn}}, n_{\text{LE}(8)}, \text{“KK-rope-chn-v1”, } 32)$$

$$\text{combined}(104) = \sigma'_{\text{ent}} \parallel \sigma'_{\text{tmp}} \parallel \sigma'_{\text{chn}} \parallel n_{\text{LE}(8)}$$

$$\text{out}(64) = \text{KK-KDF}(\text{combined}, \varepsilon_b, \text{“KK-rope-mix-v1”, } 64)$$

$$\sigma_{\text{chn}} \leftarrow \text{out}[0 \dots 31], \quad K_{\text{msg}} = \text{out}[32 \dots 63]$$

Session encode / decode:

$$\text{Encode}_{\text{session}}(P) : \quad K_{\text{msg}} \leftarrow \text{Step}(\text{Gather}()), \quad \text{return Encode}(K_{\text{msg}}, P)$$

$$\text{Decode}_{\text{session}}(\text{step}, \text{pkt}) : \quad K_{\text{msg}} \leftarrow \text{Step}(\text{step}.\varepsilon), \quad \text{return Decode}(K_{\text{msg}}, \text{pkt})$$

14 BB84 QKD Module

$$n = 4096, \quad \theta_{\text{eve}} = 0.10, \quad f_{\text{check}} = 0.25$$

Alice prepares $\forall i \in \{0, \dots, n-1\}$:

$$b_i \xleftarrow{R} \{0, 1\}, \quad \beta_i^A \xleftarrow{R} \{+, \times\}, \quad q_i = \text{encode}(b_i, \beta_i^A)$$

Eve intercepts (optional):

$$\beta_i^E \xleftarrow{R} \{+, \times\}, \quad \hat{b}_i^E = \text{measure}(q_i, \beta_i^E), \quad q_i' = \text{re-prepare}(\hat{b}_i^E, \beta_i^E)$$

Bob measures:

$$\beta_i^B \xleftarrow{R} \{+, \times\}, \quad \hat{b}_i^B = \text{measure}(q_i, \beta_i^B)$$

Sifting:

$$\mathcal{S} = \{i : \beta_i^A = \beta_i^B\}, \quad |\mathcal{S}| \geq 64$$

Error estimation:

$$\mathcal{C} \subset \mathcal{S}, \quad |\mathcal{C}| = \lfloor f_{\text{check}} \cdot |\mathcal{S}| \rfloor, \quad e = \frac{|\{i \in \mathcal{C} : b_i \neq \hat{b}_i^B\}|}{|\mathcal{C}|}$$

$$e > \theta_{\text{eve}} \implies \text{abort}$$

Privacy amplification:

$$K_{\text{qkd}} = \text{KK-KDF}((b_i)_{i \in \mathcal{S} \setminus \mathcal{C}}, \text{“BB84-KK-v1”}, \text{“KK-QKD-shared-key”}, 32)$$

QKD-secured ε transport:

$$\text{mask} = \text{KK-KDF}(\text{“QKD-epsilon-transport”}, K_{\text{qkd}}, \text{“KK-QKD-epsilon-v1”}, 48)$$

$$\text{Encrypt}_{\varepsilon}(\varepsilon) = \text{ser}(\varepsilon) \oplus \text{mask}, \quad \text{Decrypt}_{\varepsilon}(c) = c \oplus \text{mask}$$

□